

National Exit Exam from Ministry of Education Questions (1-45)

The second round of National Exit Exam (February, 2024/ Yekatit, 2016)

Exit Exam Questions for Computer Science

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National Exit Exam Questions for Computer Science.

This question was brought from Ministry of Education (Addis Ababa, Ethiopia)

1. Which one is true about the following JavaScript statement? `document.bgcolor`

- a. document is property
- b. bgcolor is method
- c. document is method
- d. bgcolor is property

Answer: d. bgcolor is property

2. Which one of the following is false about break statement in C++?

- a. break statement can be used in do...while loop
- b. break statement in nested loops terminates its nearest loop
- c. break statement can be used in if... else statements assuming the if...else statements are not inside any loop
- d. break statement can be used in switch...case statements

Answer: c. break statement can be used in if... else statements assuming the if...else statements are not inside any loop

3. Which data management approach allows application programs to define and manage their own data?

- a. Manual approach
- b. Hierarchical approach
- c. Traditional file based approach
- d. Database approach

Answer: d. Database approach

4. Which one of the following is not used for the purpose of lexical analysis?

- a. Parser
- b. Lexical analyzer
- c. Lexer
- d. Scanner

Answer: a. Parser

5. Which one of the following agents works by searching and matching the rules defined by the percept and stored internal state?

- a. Goal-based agent
- b. Utility based agent
- c. Model-based reflex agent
- d. Simple reflex agent

Answer: c. Model-based reflex agent

6. Which one of the following sorting algorithms requires additional large memory space to sort data?

- a. Merge sort
- b. Quick sort
- c. Selection sort
- d. Insertion sort

Answer: a. Merge sort

7. Assume it is given that $f(n) = 2n \log_2 n$ and $g(n) = n^2$ then which asymptotic notation is correct?

- a. $f(n) = \Omega(g(n))$
- b. $f(n) = \text{little-}\omega(g(n))$
- c. $f(n) = \Theta(g(n))$
- d. $f(n) = \text{little-}o(g(n))$

Answer: d. $f(n) = \text{little-}o(g(n))$

8. Which one of the following is false about sequential and binary searching algorithms?

- a. Binary searching algorithm can work on list of data which is sorted in descending order
- b. Considering worst case analysis binary searching algorithm is faster than sequential searching on sorted data
- c. Sequential and binary searching algorithms are not equal in terms of memory efficiency
- d. Sequential searching algorithm can work on sorted and unsorted list of data

Answer: a. Binary searching algorithm can work on list of data which is sorted in descending order

9. What is the output of the following C++ fragment code?

```
goto c;  
a:  
cout << "Graduating"  
goto b;  
c:  
cout << "Year"  
b:  
goto a;  
cout << "Fourth"
```

- a. Graduating Year Fourth
- b. Year Fourth Graduating
- c. Graduating Fourth Year
- d. Year Graduating Fourth

Answer: c. Graduating Fourth Year

10. Assume you are searching for information and typed the domain

name www.mysearchexample.com on your browser having an IP address 192.108.110.1. Since Internet communication is based on an IP address, the domain name has to be translated to its corresponding IP address. Which the following is responsible for the translation?

- a. DHCP
- b. SMTP
- c. FTP
- d. DNS

Answer: d. DNS

11. Which generation's programming languages are used in Artificial Intelligence and Expert systems?

- a. Fourth generation languages
- b. Second generation languages
- c. Fifth generation languages
- d. Third generation languages

Answer: c. Fifth generation languages

12. _____ finds the shortest path to the goal based on cost.

- a. Uniform cost search
- b. Depth first search
- c. Iterative deepening
- d. Breadth first search

Answer: a. Uniform cost search

13. Which one of the following is not correct about networking models?

- a. TCP/IP model is a protocol model
- b. A reference model has no implementation specification
- c. A network designer can use either of TCP/IP or OSI, or both
- d. Open Systems Interconnection (OSI) is a protocol model

Answer: d. Open Systems Interconnection (OSI) is a protocol model

14. Which one of the following is false about style sheets?

- a. Style sheets increase accessibility
- b. Style sheets do not reduce web page file size
- c. Style sheets create consistent appearance
- d. Style sheets make it easy to maintain HTML pages

Answer: b. Style sheets do not reduce web page file size

15. Given input Alphabet $\Sigma = (a, b, c, d)$ and empty string A. Then, which one of the following is not an element of set Σ^* ?

- a. abbccdd
- b. cd
- c. b
- d. λ

Answer: a. abbccdd

16. Which one of the following is false regarding abstract classes and abstract methods?

- a. Abstract methods are defined without implementation in the superclass
- b. We can create instances of abstract classes using the new operator
- c. Implementation of abstract methods is provided by the subclasses
- d. A class that contains abstract methods must be defined as abstract

Answer: b. We can create instances of abstract classes using the new operator

17. A traditional phone call is an example of:

- a. Broadcasting
- b. Packet switching
- c. Multiplexing
- d. Circuit switching

Answer: d. Circuit switching

18. Which one of the following is true about Queue data structure?

- a. Queue data structure applies first come first served policy
- b. Queue data structure does not have efficient basic operations
- c. Queue data structures cannot be implemented using linked list
- d. Queue data structure uses last in first out rule

Answer: a. Queue data structure applies first come first served policy

19. Which one of the following is false about roles of parser in compiling process?

- a. Parser cannot perform context free syntax analysis
- b. Parser builds the parse tree
- c. Parser produces appropriate error messages
- d. Parser attempts to correct or recover few errors

Answer: a. Parser cannot perform context free syntax analysis

20. Which method can be used to divide a browser window the smaller sub-windows so that each sub-window displaying a different HTML document?

- a. Using tables
- b. Using frames
- c. Using paragraphs
- d. Using images

Answer: b. Using frames

21. Which theory provides mechanisms for classifying combinatorial problems and measuring the computational resources necessary to solve them?

- a. Computability theory
- b. Set theory
- c. Complexity theory
- d. Automata theory

Answer: c. Complexity theory

22. In which phase of the two phase locking protocol can a transaction obtain locks?

- a. Roll back phase
- b. Shrinking phase
- c. Growing phase
- d. Release phase

Answer: c. Growing phase

23. If a company wants you to develop a software for its use, and if the company wants partial functionality to be delivered to end users without unreasonable delay. Which software development process model is appropriate for such requirement

- a. Evolutionary
- b. Waterfall
- c. Spiral
- d. Incremental

Answer: d. Incremental

24. is concerned with the internal arrangement of computer hard wares such memory, central processing units and I/O units.

- a. Computer organization
- b. Computer manufacturing
- c. Computer shipping
- d. Computer architecture

Answer: a. Computer organization

25. First design the very high level structure of the system. Then gradually worked down to detailed decisions about low-level constructs. Which design approach follows the given strategy!

- a. Top down design
- b. System design
- c. Hybrid design
- d. Bottom up design

Answer: a. Top down design

26. Which one of the following is not correct?

- a. An omniscience agent takes actions for 100% success
- b. A rational agent know the result of its action, but unrealistic
- c. Rational agent act to achieve utmost success based on information
- d. Rational agent take action but not 100% sure of success

Answer: a. An omniscience agent takes actions for 100% success

27. Look at the following PHP line:

```
$list = array("outside" => "red",  
"inside" => "brown",  
"top" => "black",  
under => "blue");
```

What is the index of data value black?

- a. outside
- b. brown
- c. under
- d. top

Answer: d. top

28. Which one of the following is responsible for controlling the bits between memory and the controller with no need for constant intervention from the CPU?

- a. Memory manager
- b. Device controller
- c. Disk fragmenting
- d. Direct memory access

Answer: d. Direct memory access

29. We should not use Waterfall software development process mode in situations like:

- a. Plenty resources with required expertise are available freely
- b. Software requirements change quickly
- c. The project is short
- d. There are no ambiguous requirements

Answer: b. Software requirements change quickly

30. Which one of the following is a mechanism undertaken to save wastage of CPU time by stopping slow I/O modules?

- a. Execution
- b. Fetching
- c. Interrupt
- d. Decoding

Answer: c. Interrupt

31. Which one of the following is a server in a network from which computer can acquire an IP address?

- a. DHCP server
- b. Web server
- c. FTP server
- d. DNS server

Answer: a. DHCP server

32. Which one of the following is not correct about magnetic disk?

- a. Its minimum block is a sector
- b. Its data is organized in tracks
- c. It needs continuous power to keep bit contents
- d. Its head is stationary during read/write

Answer: c. It needs continuous power to keep bit contents

33. Assuming there is a rectangle object, what getArea, getPerimeter can represent?

- a. Property
- b. Data field
- c. Attribute
- d. Behavior

Answer: d. Behavior

34. Which one of the following is correct array initialization?

- a. `int list[] = {23, 24, 5, 7, 14};`

```
b. int list[4] = {1, 6, 3, 9};  
int arr[4] = list;  
c int list[3] = (2, 4, 6, 8, 10));  
d. int list[6];  
list[] = [12, 1, 13, 2, 3, 5];
```

Answer: a. `int list[] = {23, 24, 5, 7, 14};`

35. _____ uses the knowledge of the characteristics of relational algebra operations and the relationship between the operators to optimize the query?

- a. Heuristic approach
- b. Deterministic approach
- c. Database approach
- d. Hierarchical data model approach

Answer: a. Heuristic approach

36. _____ is the component of the operating system that is responsible to decide which process to run first among ready processes.

- a. Memory manager
- b. Scheduler
- c. Disk manager
- d. Inter process communication

Answer: b. Scheduler

37. Which one of the following is not a problem if concurrency is not controlled properly?

- a. Incorrect summary
- b. Dirty read
- c. Lost update problem
- d. Consistent database

Answer: d. Consistent database

38. Look at the following fragment code:

```
int a = 6;  
int b = 5;  
while(a > b)  
cout << "Computer Science program" << endl;
```

Which algorithm property is not satisfied by the above code algorithm?

- a. Definiteness
- b. Finiteness
- c. Feasibility
- d. Sequential

Answer: b. Finiteness

39. Which of the following is concerned with the meaning of a sentence in knowledge representation?

- a. Structure
- b. Computational aspect

- c. Semantics
- d. Syntax

Answer: c. Semantics

40. Which one of the following is not a condition for two operations in a schedule to be said to conflict?

- a. If they access the same item X
- b. If they belong to different transactions
- c. If Write_item (X) is at least one of the operations
- d. If they belong to the same transactions

Answer: d. If they belong to the same transactions

41. Assume cities in Ethiopia are represented by vertices of a graph edges are weighted and directed. Weights of edges show distance connecting two cities. Which algorithm is appropriate to find shortest distance of all cities from Addis Ababa?

- a. Dijkstra's algorithm
- b. Prim's algorithm
- c. Kruskal's algorithm
- d. Huffman encoding algorithm

Answer: a. Dijkstra's algorithm

42. Which operator is used to create an array?

- a. static
- b. declare
- c. construct
- d. new

Answer: d. new

43. Which one of the following is false about client-side scripting?

- a. Script languages have limited capabilities
- b. Script languages are not full-featured
- c. Script languages can access client's hardware
- d. Script languages are embedded as plain text, and interpreted by applications

Answer: c. Script languages can access client's hardware

44. Which one of the following is false regarding Exceptions in Java?

- a. We cannot create our own exception classes by extending Exception class
- b. All Java exception classes inherit directly or indirectly from Throwable class
- c. RuntimeException class describes programming errors such as bad casting and numeric error
- d. Exception class describes errors caused by your program and external circumstances

Answer: a. We cannot create our own exception classes by extending Exception class

45. Which one of the following is an attack detection mechanism?

- a. Password
- b. Physical control

- c. Encryption
- d. Audit log

Answer: d. Audit log

THANK YOU!